

Low-Temperature Electronics for Space

Cryogenic robotics, low-temperature electronics and high-temperature superconductivity need collaboration to shape the future of lunar research and exploration

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What is the temperature on other planets? -150°C at the orbit of Jupiter, -230°C at the orbit of Pluto and below -55°C to -65°C at moon, Mars and asteroids. At such low temperatures, conventional-temperature electronics is not of help.

So researchers are pioneering a technology called 'low-temperature electronics' or 'cryoelectronics'. Cryoelectronic materials with high carrier mobility and reliability might provide a better solution for circuits to be used in low temperatures.

What is cryoelectronics?

'Cryo' is derived from Greek word 'kryo' that means 'icy cold' or 'frost'. Cryogenic temperature ranges from -150°C to -273°C . Cryoelectronics deals with the production of low-temperature electronic devices and utilisation of low-temperature phenomenon. Examples of these devices include complementary metal-oxide semiconductor (CMOS) diodes and field-effect transistors.

Why traditional electronics cannot be used in space?

Electronics plays an important role in spacecrafts. It is used in sensors, cameras, wireless data transmission, etc. Conventional-temperature electronics works between $-55^{\circ}\text{C}/-65^{\circ}\text{C}$ and 125°C . Low-temperature electronics, on the other hand, works at temperatures ranging from -150°C to -273°C (refer Fig. 1).

In military and scientific aircrafts, many sensors such as X-ray detectors are used for astronomical observation or surveillance. These

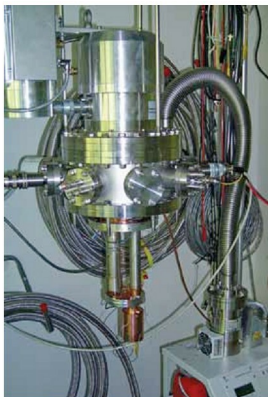
must operate at a low temperature. Cosmic background explorer, infrared telescope in space and infrared space observatory are some of the

devices which make use of cryoelectronics-based electronic components.

Conventional-temperature electronic circuits require a thermal source for low-temperature applications. In many situations, such techniques are undesirable or impractical. Passive techniques have limited lifespans. Active technologies, on the other hand, require additional power and sub-systems. This makes electronic circuits bulky, heavy and complex.

Cryoelectronics with superconductors

Superconductivity is a state of matter. The thermal, electric and magnetic properties of many chemical elements, compounds and alloys become strikingly



Testing equipment for low-temperature electronics

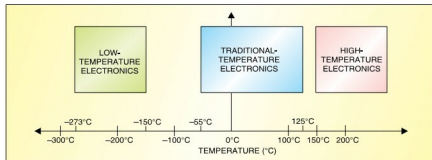


Fig. 1: Temperature range for different temperature-based electronics